

Scanners for your shoes

Now it's shoe time. It's not a long way that your shoes will also be scanned at the airport for security.

A Mixed Bag

Motorola Atrix is a 4G Android phone which also possesses a unique feature of a laptop-shaped dock

4 and the latest OMAP 5. ARM's NEON processor is the media-processing engine, and TI has their own programmable DSP with support for encoding/decoding and full HD playback. When it comes to DSP, TI has a distinct advantage over other SoC manufacturers with a lot of experience in this realm, particularly in terms of quality.

The OMAP 5 series is a lot more powerful, with support for goodies like 3D video recording, 8 GB of DDR3 memory, and up to 3 USB 2.0 ports and 1 SATA 2.0 controller. This SoC is also intended for tablets and low-power notebooks.

Experts feel that given TI's expertise with hardware decoding and DSP, OMAP4s should be better than other dual-core SoC on related tasks. The main disadvantage for TI is their design doesn't include a baseband modem, necessitating an additional chip. However, they use their own wireless solution, and this module contains FM transmitter, Bluetooth and Wi-Fi. The real question here is, how many vendors will TI garner?

ST-Ericsson

Besides manufacturing SoC, ST-Ericsson is a major player in the wireless platform arena and is the largest provider of baseband modems and other wireless components like Bluetooth and Wi-Fi chips to major handset providers, including Sony Ericsson and even Nokia.

The U8500 is their most prominent SoC, and is based on the Cortex-A9 design. As with other SoC designs – and unlike Snapdragon – baseband modem, Bluetooth, Wi-Fi and GPS functionality on the U8500 is not integrated. ST-Ericsson is currently working hard to ensure their SoC are optimised for Windows Phone 7, and ST-Ericsson has predicted a sudden surge in interest on what is a promising platform – certainly the best MS has come out with by far.

Nokia, (ST-Ericsson's biggest client), will make phones that run WP7, so we're going to see Nokia buying a lot of SoC parts. And here lies the problem, for Qualcomm is the only SoC vendor who is MS-certified to build SoC for WP7, and they've already got products ready, therefore ST-Ericsson has some catching up to do.

At the MWC 2011 a new SoC called the Nova A9600 was announced, Admit-

Parameters	ARM Cortex A8	ARM Cortex A9	ARM Cortex A15
Architecture	RISC-based	RISC-based	RISC-based
CPU Speeds	Up to 1.2 GHz	Up to 2 GHz (per core)	Up to 2.5 GHz (per core)
Multiprocessor support	N	Y	Y
Supported no. of CPU cores	1 core	up to 4 cores	up to 8 cores
Instruction set	ARMv7A	ARMv7A	ARMv7A
SoCs based on this architecture	Qualcomm Snapdragon QSD8x50, MSM7x30, MSM8255, TI OMAP 3xxx series, Apple A4, Samsung Hummingbird S5PC100, S5PC110	NVIDIA Tegra 2, TI OMAP44x0, ST-Ericsson U9500, U9450, U8500, U5500, Samsung Orion, Apple A5	Qualcomm Snapdragon MSM8270, MSM8960, Texas Instruments OMAP5, Samsung Eagle, ST-Ericsson NOVA U9600, NVIDIA Wayne

tedly, this SoC is for tablets. The GPU is very new, in fact it hasn't been announced yet, and is codenamed "Rouge".

The Nova A9500 is a mid-range SoC, based on a 45nm process. It supports full 1080p HD video decoding, and a camera of up to 20-megapixels. Combined with ST-Ericsson's own Thor M5730 baseband and modem, the combination of application processors and communications modem is being called the NovaThor U9500. Then there is the Nova A9540. This is a 32nm SoC, and boasts of a dual core ARM Cortex-A9 CPU clocked at up to 1.8 GHz. The graphics system is claimed to be four times more powerful than the U8500.

Intel

While everyone is jumping on to the ARM bandwagon, Intel is choosing to stick with x86, and are using their Atom processor to combat the likes of the ARM Cortex-A9. Intel hopes to use Atom for smartphones and tablets in the near future. Their biggest advantage seems to be their fabrication process, which is probably the most advanced in the world.

Intel's Medfield platform will support Android. This is obvious, since Android is open source, and can easily be ported to x86. The uphill task here is garnering support from developers to build software for an x86 version of Android. Medfield will also have support for platforms such as Moblin (MeeGo) and Ubuntu Mobile. We already know Intel's Oak Trail is slated to appear in tablets, by March end this year. Again, Oak Trail is an SoC based on the Atom CPU, and Intel has been spending the last year or so tweaking performance and battery life for Windows 7. While details on Oak Trail's hardware are a bit sketchy, the Z670 Atom processor running at 1.5 GHz or 1.66 GHz could be the CPU, and this platform should handle full HD 1080p content well.

What's next?

ARM seems to be the future. Proof? In January this year, MS announced the next edition of Windows will support ARM as well as x86. Steven Sinofsky, president of the Windows and Windows Live division of MS, said this move "was meant to show the flexibility and resiliency of Windows, through the power of software, and a commitment to world class engineering".

From an inconspicuous component, to spearheading marketing strategy SoC has come a long way. However, fast hardware is no substitute for a good interface, or a usable keypad. Therefore, it's important to get the basics right. We're still performance junkies, of course, but the mobile platform demands more than just dual cores and 1080p HD videos on phones.

The future is visual and interactive computing, and increased convergence is the key. How about being able to download any sort of file on your phone and work on it, then sending to your computer and a speed of 100 MBps? What if, at some point, we could run the same software on all our devices where syncing and switching between devices would be a non-existent chore? Think code reusability.

Imagine your desktop unobtrusively encoding a video file present on your cellphone, while you're lying in bed with it, playing a game. Or your phone instantly translating foreign text on a billboard, captured by its camera, into your language – in real time.

Power deludes is a time-proven adage, and we're hoping marketing hype focuses on improving devices, and not just their underlying hardware. We want to be able to do more on our handhelds and tablets, but we don't want, or for that matter, need, another Gigahertz war. 